

Experiment No: 9

Title

Write a Python Program to Implement the Travelling Salesman Problem (TSP)

Aim

To develop a Python program to solve the **Travelling Salesman Problem (TSP)** using the **Nearest Neighbour** approach and determine the shortest tour that visits all cities exactly once and returns to the starting city.

Objective

To implement the Travelling Salesman Problem by calculating the distances between cities and selecting the nearest unvisited city until all cities are visited.

Algorithm

Step 1: Start the program.

Step 2: Read the coordinates of all cities.

Step 3: Construct the distance matrix using Euclidean distance.

Step 4: Select the first city as the starting city.

Step 5: Find the nearest unvisited city.

Step 6: Add the nearest city to the tour and update the total distance.

Step 7: Repeat Steps 5 and 6 until all cities are visited.

Step 8: Return to the starting city.

Step 9: Display the tour and total distance.

Step 10: Stop the program.

Program

```
import numpy as np
```

```
def tsp(cities):
```

```

n = len(cities)

# Create Distance Matrix
distance = np.zeros((n, n))

for i in range(n):
    for j in range(n):
        if i != j:
            distance[i][j] = np.linalg.norm(cities[i] - cities[j])

visited = [False] * n
visited[0] = True

tour = [0]
total_distance = 0
current = 0

while len(tour) < n:

    nearest = -1
    minimum = float('inf')

    for city in range(n):

        if not visited[city] and distance[current][city] < minimum:
            minimum = distance[current][city]
            nearest = city

    tour.append(nearest)

```

```
visited[nearest] = True
total_distance += minimum
current = nearest
```

```
total_distance += distance[current][0]
tour.append(0)
```

```
return tour, total_distance
```

```
# Main Program
```

```
n = int(input("Enter Number of Cities: "))
```

```
cities = []
```

```
for i in range(n):
    x = float(input(f'Enter x-coordinate of City {i}: '))
    y = float(input(f'Enter y-coordinate of City {i}: '))
    cities.append(np.array([x, y]))
```

```
tour, distance = tsp(cities)
```

```
print("\nOptimal Tour")
```

```
print(" -> ".join(map(str, tour)))
```

```
print("Total Distance =", round(distance,2))
```

Sample Input

Enter Number of Cities: 4

City 0 : (0,0)

City 1 : (2,3)

City 2 : (5,2)

City 3 : (6,6)

Sample Output

```
Enter Number of Cities: 4
Enter x-coordinate of City 0: 0
Enter y-coordinate of City 0: 0
Enter x-coordinate of City 1: 2
Enter y-coordinate of City 1: 3
Enter x-coordinate of City 2: 5
Enter y-coordinate of City 2: 2
Enter x-coordinate of City 3: 6
Enter y-coordinate of City 3: 6
```

```
Optimal Tour
0 -> 1 -> 2 -> 3 -> 0
Total Distance = 19.38
```

Result: Thus the Program was executed and the output was found to be Correct.

